Answer key - 1 of 2

Exercise	Values	Exercise	Values
1	A=12, B=1	36	A=8, B=9
2	A=1, B=4	37	A=6, B=17
3	A=12, B=4	38	A=13, B=11
4	A=3, B=16	39	A=6, B=12
5	A=9, B=10	40	A=15, B=19
6	A=16, B=9	41	A=6, B=3
7	A=7, B=14	42	A=8, B=4
8	A=19, B=13	43	A=4, B=12
9	A=15, B=1	44	A=19, B=20
10	A=3, B=1	45	A=16, B=18
11	A=9, B=18	46	A=15, B=3
12	A=3, B=14	47	A=9, B=3
13	A=2, B=1	48	A=2, B=6, C=4
14	A=17, B=20	49	A=8, B=16, C=4
15	A=8, B=2	50	A=13, B=2, C=7
16	A=13, B=5	51	A=8, B=6, C=16
17	A=19, B=9	52	A=7, B=12, C=4
18	A=20, B=5	53	A=2, B=1, C=2
19	A=4, B=8	54	A=4, B=5, C=9
20	A=16, B=13	55	A=8, B=3, C=1
21	A=20, B=11	56	A=20, B=8, C=2
22	A=16, B=15	57	A=12, B=3, C=6
23	A=15, B=5	58	A=13, B=19, C=9
24	A=4, B=18	59	A=6, B=18, C=5
25	A=13, B=6	60	A=2, B=2, C=4
26	A=2, B=8	61	A=6, B=12, C=9
27	A=5, B=20	62	A=12, B=4, C=8
28	A=16, B=15	63	A=8, B=11, C=4
29	A=8, B=19	64	A=2, B=18, C=8
30	A=5, B=3	65	A=12, B=4, C=8
31	A=12, B=4	66	A=14, B=10, C=18
32	A=5, B=8	67	A=14, B=5, C=12
33	A=9, B=5	68	A=10, B=8, C=5
34	A=12, B=4	69	A=2, B=8, C=9
35	A=10, B=5	70	A=15, B=1, C=16

Answer key - 2 of 2

Exercise	Values	Exercise	Values
71	A=1, B=2, C=4	106	A=15, B=3, C=7, D=4
72	A=5, B=1, C=4	107	A=8, B=10, C=16, D=6
73	A=9, B=6, C=3	108	A=15, B=8, C=10, D=5
74	A=9, B=10, C=4	109	A=17, B=15, C=5, D=7
75	A=11, B=17, C=19	110	A=4, B=16, C=8, D=7
76	A=8, B=2, C=5	111	A=9, B=2, C=9, D=10
77	A=8, B=4, C=6	112	A=17, B=2, C=4, D=13
78	A=12, B=2, C=10	113	A=13, B=4, C=11, D=1
79	A=12, B=7, C=3	114	A=13, B=8, C=9, D=4
80	A=4, B=2, C=1	115	A=4, B=14, C=20, D=6
81	A=8, B=4, C=12	116	A=2, B=7, C=8, D=8
82	A=13, B=15, C=9	117	A=11, B=6, C=19, D=5
83	A=5, B=4, C=10	118	A=3, B=9, C=4, D=12
84	A=10, B=5, C=8	119	A=9, B=6, C=3, D=5
85	A=4, B=2, C=1	120	A=6, B=2, C=2, D=3
86	A=14, B=8, C=6	121	A=3, B=12, C=4, D=7
87	A=11, B=20, C=5	122	A=2, B=9, C=11, D=6
88	A=3, B=12, C=15	123	A=7, B=13, C=10, D=3
89	A=4, B=2, C=4	124	A=4, B=9, C=11, D=3
90	A=19, B=13, C=8	125	A=5, B=11, C=15, D=7
91	A=10, B=18, C=12	126	A=9, B=6, C=9, D=3
92	A=10, B=16, C=4	127	A=9, B=12, C=3, D=8
93	A=8, B=2, C=8	128	A=7, B=1, C=4, D=2
94	A=12, B=9, C=3, D=9	129	A=11, B=5, C=10, D=20
95	A=14, B=6, C=5, D=20	130	A=11, B=9, C=8, D=3
96	A=1, B=2, C=2, D=3	131	A=8, B=4, C=4, D=8
97	A=2, B=4, C=12, D=6	132	A=3, B=13, C=5, D=12
98	A=12, B=10, C=4, D=6	133	A=9, B=9, C=3, D=6
99	A=3, B=13, C=10, D=2	134	A=3, B=6, C=12, D=9
100	A=3, B=6, C=12, D=6	135	A=4, B=9, C=8, D=7
101	A=7, B=5, C=2, D=4	136	A=7, B=7, C=3, D=5
102	A=5, B=14, C=20, D=20	137	A=1, B=6, C=5, D=2
103	A=8, B=9, C=2, D=10	138	A=3, B=10, C=12, D=7
104	A=17, B=7, C=5, D=4	139	A=14, B=3, C=11, D=1
105	A=6, B=2, C=3, D=9	140	A=5, B=15, C=11, D=7

Solution - Exercise 1

(1) A - 11 = B **A = 12**

(2) 7 + B = 8 **B = 1**

Use equation 2: B = 1. Substitute into equation 1: A - 11 = 1. This gives A = 12; substitute back to obtain all the values shown.

Check every equation:

12 - 11 = 1 [both sides = 1]

7 + 1 = 8 [both sides = 8]

Solution - Exercise 2

(1) B ÷ 4 = A **A = 1**

(2) B - A = 3 **B = 4**

Use equation 1: B = 4 × A. Substitute into equation 2: 3 × A = 3. This gives A = 1; substitute back to obtain all the values shown.

Check every equation:

4 ÷ 4 = 1 [both sides = 1]

4 - 1 = 3 [both sides = 3]

Solution - Exercise 3

(1) B + A = 16 **A = 12**

(2) 3 × B = A **B = 4**

Use equation 2: A = 3 × B. Substitute into equation 1: 4 × B = 16. This gives B = 4; substitute back to obtain all the values shown.

Check every equation:

4 + 12 = 16 [both sides = 16]

3 × 4 = 12 [both sides = 12]

Solution - Exercise 4

(1) B - A = 13 **A = 3**

(2) B + A = 19 **B = 16**

Use equation 2: B = 19 - A. Substitute into equation 1: 19 - 2 × A = 13. This gives A = 3; substitute back to obtain all the values shown.

Check every equation:

16 - 3 = 13 [both sides = 13]

16 + 3 = 19 [both sides = 19]

Solution - Exercise 5

(1) A + 1 = B **A = 9**

(2) 2 × A = B + 8 **B = 10**

Use equation 1: B = A + 1. Substitute into equation 2: 2 × A = A + 9. This gives A = 9; substitute back to obtain all the values shown.

Check every equation:

9 + 1 = 10 [both sides = 10]

2 × 9 = 10 + 8 [both sides = 18]

Solution - Exercise 6

(1) 4 × B = 36 **A = 16**

(2) B + A = 25 **B = 9**

Use equation 1: B = 9. Substitute into equation 2: A + 9 = 25. This gives A = 16; substitute back to obtain all the values shown.

Check every equation:

4 × 9 = 36 [both sides = 36]

9 + 16 = 25 [both sides = 25]

Solution - Exercise 7

(1) B ÷ 2 = A **A = 7**

(2) 21 - B = A **B = 14**

Use equation 1: B = 2 × A. Substitute into equation 2: 21 - 2 × A = A. This gives A = 7; substitute back to obtain all the values shown.

Check every equation:

14 ÷ 2 = 7 [both sides = 7]

21 - 14 = 7 [both sides = 7]

Solution - Exercise 8

(1) A = B + 6 **A = 19**

(2) A + 3 × B = 58 **B = 13**

Use equation 1: A = B + 6. Substitute into equation 2: 4 × B + 6 = 58. This gives B = 13; substitute back to obtain all the values shown.

Check every equation:

19 = 13 + 6 [both sides = 19]

19 + 3 × 13 = 58 [both sides = 58]

Solution - Exercise 9

(1) 1 + A = 16 **A = 15**

(2) B + 14 = A **B = 1**

Use equation 1: A = 15. Substitute into equation 2: B + 14 = 15. This gives B = 1; substitute back to obtain all the values shown.

Check every equation:

1 + 15 = 16 [both sides = 16]

1 + 14 = 15 [both sides = 15]

Solution - Exercise 10

(1) A - B = 2 **A = 3**

(2) A ÷ 3 = B **B = 1**

Use equation 2: A = 3 × B. Substitute into equation 1: 2 × B = 2. This gives B = 1; substitute back to obtain all the values shown.

Check every equation:

3 - 1 = 2 [both sides = 2]

3 ÷ 3 = 1 [both sides = 1]

Solution - Exercise 11

(1) 2 × A = B **A = 9**

(2) A + B = 27 **B = 18**

Use equation 1: B = 2 × A. Substitute into equation 2: 3 × A = 27. This gives A = 9; substitute back to obtain all the values shown.

Check every equation:

2 × 9 = 18 [both sides = 18]

9 + 18 = 27 [both sides = 27]

Solution - Exercise 12

(1) B - A = 11 **A = 3**

(2) B + A = 17 **B = 14**

Use equation 2: B = 17 - A. Substitute into equation 1: 17 - 2 × A = 11. This gives A = 3; substitute back to obtain all the values shown.

Check every equation:

14 - 3 = 11 [both sides = 11]

14 + 3 = 17 [both sides = 17]

Solution - Exercise 13

(1) $3 \times B = A + 1$ **A = 2**

(2) $B + 1 = A$ **B = 1**

Use equation 2: A = B + 1. Substitute into equation 1: 3 × B = B + 2. This gives B = 1; substitute back to obtain all the values shown.

Check every equation:

3 × 1 = 2 + 1 [both sides = 3]

1 + 1 = 2 [both sides = 2]

Solution - Exercise 14

(1) $B + A = 37$ **A = 17**

(2) $3 \times B = 60$ **B = 20**

Use equation 2: B = 20. Substitute into equation 1: A + 20 = 37. This gives A = 17; substitute back to obtain all the values shown.

Check every equation:

20 + 17 = 37 [both sides = 37]

3 × 20 = 60 [both sides = 60]

Solution - Exercise 15

(1) $10 - A = B$ **A = 8**

(2) $A \div 4 = B$ **B = 2**

Use equation 2: A = 4 × B. Substitute into equation 1: 10 - 4 × B = B. This gives B = 2; substitute back to obtain all the values shown.

Check every equation:

10 - 8 = 2 [both sides = 2]

8 ÷ 4 = 2 [both sides = 2]

Solution - Exercise 16

(1) $A = B + 8$ **A = 13**

(2) $A + 3 \times B = 28$ **B = 5**

Use equation 1: A = B + 8. Substitute into equation 2: 4 × B + 8 = 28. This gives B = 5; substitute back to obtain all the values shown.

Check every equation:

13 = 5 + 8 [both sides = 13]

13 + 3 × 5 = 28 [both sides = 28]

Solution - Exercise 17

(1) $B + 10 = A$ **A = 19**

(2) $8 + A = 27$ **B = 9**

Use equation 2: A = 19. Substitute into equation 1: B + 10 = 19. This gives B = 9; substitute back to obtain all the values shown.

Check every equation:

9 + 10 = 19 [both sides = 19]

8 + 19 = 27 [both sides = 27]

Solution - Exercise 18

(1) $A \div 4 = B$ **A = 20**

(2) $A - B = 15$ **B = 5**

Use equation 1: A = 4 × B. Substitute into equation 2: 3 × B = 15. This gives B = 5; substitute back to obtain all the values shown.

Check every equation:

20 ÷ 4 = 5 [both sides = 5]

20 - 5 = 15 [both sides = 15]

Solution - Exercise 19

(1) $A + B = 12$ **A = 4**

(2) $2 \times A = B$ **B = 8**

Use equation 2: B = 2 × A. Substitute into equation 1: 3 × A = 12. This gives A = 4; substitute back to obtain all the values shown.

Check every equation:

4 + 8 = 12 [both sides = 12]

2 × 4 = 8 [both sides = 8]

Solution - Exercise 20

(1) $A + B = 29$ **A = 16**

(2) $A - B = 3$ **B = 13**

Use equation 2: B = A - 3. Substitute into equation 1: 2 × A - 3 = 29. This gives A = 16; substitute back to obtain all the values shown.

Check every equation:

16 + 13 = 29 [both sides = 29]

16 - 13 = 3 [both sides = 3]

Solution - Exercise 21

(1) $B + 9 = A$ **A = 20**

(2) $2 \times B = A + 2$ **B = 11**

Use equation 1: A = B + 9. Substitute into equation 2: 2 × B = B + 11. This gives B = 11; substitute back to obtain all the values shown.

Check every equation:

11 + 9 = 20 [both sides = 20]

2 × 11 = 20 + 2 [both sides = 22]

Solution - Exercise 22

(1) $4 \times B = 60$ **A = 16**

(2) $B + A = 31$ **B = 15**

Use equation 1: B = 15. Substitute into equation 2: A + 15 = 31. This gives A = 16; substitute back to obtain all the values shown.

Check every equation:

4 × 15 = 60 [both sides = 60]

15 + 16 = 31 [both sides = 31]

Solution - Exercise 23

(1) $20 - A = B$ **A = 15**

(2) $A \div 3 = B$ **B = 5**

Use equation 2: A = 3 × B. Substitute into equation 1: 20 - 3 × B = B. This gives B = 5; substitute back to obtain all the values shown.

Check every equation:

20 - 15 = 5 [both sides = 5]

15 ÷ 3 = 5 [both sides = 5]

Solution - Exercise 24

(1) $B = A + 14$ **A = 4**

(2) $B + 3 \times A = 30$ **B = 18**

Use equation 1: B = A + 14. Substitute into equation 2: 4 × A + 14 = 30. This gives A = 4; substitute back to obtain all the values shown.

Check every equation:

18 = 4 + 14 [both sides = 18]

18 + 3 × 4 = 30 [both sides = 30]

Solution - Exercise 25

(1) A - 7 = B **A = 13**

(2) 4 + B = 10 **B = 6**

Use equation 2: B = 6. Substitute into equation 1: A - 7 = 6. This gives A = 13; substitute back to obtain all the values shown.

Check every equation:

13 - 7 = 6 [both sides = 6]

4 + 6 = 10 [both sides = 10]

Solution - Exercise 26

(1) B ÷ 4 = A **A = 2**

(2) B - A = 6 **B = 8**

Use equation 1: B = 4 × A. Substitute into equation 2: 3 × A = 6. This gives A = 2; substitute back to obtain all the values shown.

Check every equation:

8 ÷ 4 = 2 [both sides = 2]

8 - 2 = 6 [both sides = 6]

Solution - Exercise 27

(1) A + B = 25 **A = 5**

(2) 4 × A = B **B = 20**

Use equation 2: B = 4 × A. Substitute into equation 1: 5 × A = 25. This gives A = 5; substitute back to obtain all the values shown.

Check every equation:

5 + 20 = 25 [both sides = 25]

4 × 5 = 20 [both sides = 20]

Solution - Exercise 28

(1) $A + B = 31$ **A = 16**

(2) $A - B = 1$ **B = 15**

Use equation 2: B = A - 1. Substitute into equation 1: 2 × A - 1 = 31. This gives A = 16; substitute back to obtain all the values shown.

Check every equation:

16 + 15 = 31 [both sides = 31]

16 - 15 = 1 [both sides = 1]

Solution - Exercise 29

(1) $3 \times A = B + 5$ **A = 8**

(2) $A + 11 = B$ **B = 19**

Use equation 2: B = A + 11. Substitute into equation 1: 3 × A = A + 16. This gives A = 8; substitute back to obtain all the values shown.

Check every equation:

3 × 8 = 19 + 5 [both sides = 24]

8 + 11 = 19 [both sides = 19]

Solution - Exercise 30

(1) $A + B = 8$ **A = 5**

(2) $4 \times A = 20$ **B = 3**

Use equation 2: A = 5. Substitute into equation 1: B + 5 = 8. This gives B = 3; substitute back to obtain all the values shown.

Check every equation:

5 + 3 = 8 [both sides = 8]

4 × 5 = 20 [both sides = 20]

Solution - Exercise 31

(1) 16 - A = B **A = 12**

(2) A ÷ 3 = B **B = 4**

Use equation 2: A = 3 × B. Substitute into equation 1: 16 - 3 × B = B. This gives B = 4; substitute back to obtain all the values shown.

Check every equation:

16 - 12 = 4 [both sides = 4]

12 ÷ 3 = 4 [both sides = 4]

Solution - Exercise 32

(1) B = A + 3 **A = 5**

(2) B + 2 × A = 18 **B = 8**

Use equation 1: B = A + 3. Substitute into equation 2: 3 × A + 3 = 18. This gives A = 5; substitute back to obtain all the values shown.

Check every equation:

8 = 5 + 3 [both sides = 8]

8 + 2 × 5 = 18 [both sides = 18]

Solution - Exercise 33

(1) 4 + A = 13 **A = 9**

(2) B + 4 = A **B = 5**

Use equation 1: A = 9. Substitute into equation 2: B + 4 = 9. This gives B = 5; substitute back to obtain all the values shown.

Check every equation:

4 + 9 = 13 [both sides = 13]

5 + 4 = 9 [both sides = 9]

Solution - Exercise 34

(1) A - B = 8 **A = 12**

(2) A ÷ 3 = B **B = 4**

Use equation 2: A = 3 × B. Substitute into equation 1: 2 × B = 8. This gives B = 4; substitute back to obtain all the values shown.

Check every equation:

12 - 4 = 8 [both sides = 8]

12 ÷ 3 = 4 [both sides = 4]

Solution - Exercise 35

(1) 2 × B = A **A = 10**

(2) B + A = 15 **B = 5**

Use equation 1: A = 2 × B. Substitute into equation 2: 3 × B = 15. This gives B = 5; substitute back to obtain all the values shown.

Check every equation:

2 × 5 = 10 [both sides = 10]

5 + 10 = 15 [both sides = 15]

Solution - Exercise 36

(1) B - A = 1 **A = 8**

(2) B + A = 17 **B = 9**

Use equation 1: B = A + 1. Substitute into equation 2: 2 × A + 1 = 17. This gives A = 8; substitute back to obtain all the values shown.

Check every equation:

9 - 8 = 1 [both sides = 1]

9 + 8 = 17 [both sides = 17]

Solution - Exercise 37

(1) $4 \times A = B + 7$ **A = 6**

(2) $A + 11 = B$ **B = 17**

Use equation 2: B = A + 11. Substitute into equation 1: 4 × A = A + 18. This gives A = 6; substitute back to obtain all the values shown.

Check every equation:

4 × 6 = 17 + 7 [both sides = 24]

6 + 11 = 17 [both sides = 17]

Solution - Exercise 38

(1) $A + B = 24$ **A = 13**

(2) $2 \times A = 26$ **B = 11**

Use equation 2: A = 13. Substitute into equation 1: B + 13 = 24. This gives B = 11; substitute back to obtain all the values shown.

Check every equation:

13 + 11 = 24 [both sides = 24]

2 × 13 = 26 [both sides = 26]

Solution - Exercise 39

(1) $18 - B = A$ **A = 6**

(2) $B \div 2 = A$ **B = 12**

Use equation 2: B = 2 × A. Substitute into equation 1: 18 - 2 × A = A. This gives A = 6; substitute back to obtain all the values shown.

Check every equation:

18 - 12 = 6 [both sides = 6]

12 ÷ 2 = 6 [both sides = 6]

Solution - Exercise 40

(1) $B + 4 \times A = 79$ **A = 15**
(2) $B = A + 4$ **B = 19**

Use equation 2: B = A + 4. Substitute into equation 1: 5 × A + 4 = 79. This gives A = 15; substitute back to obtain all the values shown.
Check every equation:
19 + 4 × 15 = 79 [both sides = 79]
19 = 15 + 4 [both sides = 19]

Solution - Exercise 41

(1) $1 + B = 4$ **A = 6**
(2) $A - 3 = B$ **B = 3**

Use equation 1: B = 3. Substitute into equation 2: A - 3 = 3. This gives A = 6; substitute back to obtain all the values shown.
Check every equation:
1 + 3 = 4 [both sides = 4]
6 - 3 = 3 [both sides = 3]

Solution - Exercise 42

(1) $A - B = 4$ **A = 8**
(2) $A \div 2 = B$ **B = 4**

Use equation 2: A = 2 × B. Substitute into equation 1: B = 4. This gives B = 4; substitute back to obtain all the values shown.
Check every equation:
8 - 4 = 4 [both sides = 4]
8 ÷ 2 = 4 [both sides = 4]

Solution - Exercise 43

(1) $A + B = 16$ **A = 4**
(2) $3 \times A = B$ **B = 12**

Use equation 2: B = 3 × A. Substitute into equation 1: 4 × A = 16. This gives A = 4; substitute back to obtain all the values shown.
Check every equation:
4 + 12 = 16 [both sides = 16]
3 × 4 = 12 [both sides = 12]

Solution - Exercise 44

(1) $B + A = 39$ **A = 19**
(2) $B - A = 1$ **B = 20**

Use equation 2: B = A + 1. Substitute into equation 1: 2 × A + 1 = 39. This gives A = 19; substitute back to obtain all the values shown.
Check every equation:
20 + 19 = 39 [both sides = 39]
20 - 19 = 1 [both sides = 1]

Solution - Exercise 45

(1) $A + 2 = B$ **A = 16**
(2) $2 \times A = B + 14$ **B = 18**

Use equation 1: B = A + 2. Substitute into equation 2: 2 × A = A + 16. This gives A = 16; substitute back to obtain all the values shown.
Check every equation:
16 + 2 = 18 [both sides = 18]
2 × 16 = 18 + 14 [both sides = 32]

Solution - Exercise 46

(1) $A + B = 18$ **A = 15**

(2) $3 \times A = 45$ **B = 3**

Use equation 2: A = 15. Substitute into equation 1: B + 15 = 18. This gives B = 3; substitute back to obtain all the values shown.

Check every equation:

15 + 3 = 18 [both sides = 18]

3 × 15 = 45 [both sides = 45]

Solution - Exercise 47

(1) $12 - A = B$ **A = 9**

(2) $A \div 3 = B$ **B = 3**

Use equation 2: A = 3 × B. Substitute into equation 1: 12 - 3 × B = B. This gives B = 3; substitute back to obtain all the values shown.

Check every equation:

12 - 9 = 3 [both sides = 3]

9 ÷ 3 = 3 [both sides = 3]

Solution - Exercise 48

(1) $2 \times A = C$ **A = 2**

(2) $1 \times C + 1 \times A = B$ **B = 6**

(3) $C + A = 6$ **C = 4**

Use equations 1, 3: A = 2; C = 4. Substitute into equation 2: 6 = B. This gives B = 6; substitute back to obtain all the values shown.

Check every equation:

2 × 2 = 4 [both sides = 4]

1 × 4 + 1 × 2 = 6 [both sides = 6]

4 + 2 = 6 [both sides = 6]

Solution - Exercise 49

(1)	20 - B = C	**A = 8**
(2)	2 × C = A	**B = 16**
(3)	B ÷ 4 = C	**C = 4**

Use equations 1, 3: B = 16; C = 4. Substitute into equation 2: 8 = A. This gives A = 8; substitute back to obtain all the values shown.

Check every equation:

20 - 16 = 4 [both sides = 4]
2 × 4 = 8 [both sides = 8]
16 ÷ 4 = 4 [both sides = 4]

Solution - Exercise 50

(1)	A = C + 6	**A = 13**
(2)	A + C + B = 22	**B = 2**
(3)	C = B + 5	**C = 7**

Use equations 1, 3: A = C + 6; B = C - 5. Substitute into equation 2: 3 × C + 1 = 22. This gives C = 7; substitute back to obtain all the values shown.

Check every equation:

13 = 7 + 6 [both sides = 13]
13 + 7 + 2 = 22 [both sides = 22]
7 = 2 + 5 [both sides = 7]

Solution - Exercise 51

(1)	A + C = 24	**A = 8**
(2)	A + B = 14	**B = 6**
(3)	B + C = 22	**C = 16**

Use equations 2, 3: B = 14 - A; C = A + 8. Substitute into equation 1: 2 × A + 8 = 24. This gives A = 8; substitute back to obtain all the values shown.

Check every equation:

8 + 16 = 24 [both sides = 24]
8 + 6 = 14 [both sides = 14]
6 + 16 = 22 [both sides = 22]

Solution - Exercise 52

(1) $B = 3 \times C$ **A = 7**
(2) $C + A = 11$ **B = 12**
(3) $A = B - 5$ **C = 4**

Use equations 1, 3: $A = 3 \times C - 5$; $B = 3 \times C$. Substitute into equation 2: $4 \times C - 5 = 11$. This gives C = 4; substitute back to obtain all the values shown.
Check every equation:
$12 = 3 \times 4$ [both sides = 12]
$4 + 7 = 11$ [both sides = 11]
$7 = 12 - 5$ [both sides = 7]

Solution - Exercise 53

(1) $C + A = 4$ **A = 2**
(2) $A = B + 1$ **B = 1**
(3) $B = C \div 2$ **C = 2**

Use equations 2, 3: $A = B + 1$; $C = 2 \times B$. Substitute into equation 1: $3 \times B + 1 = 4$. This gives B = 1; substitute back to obtain all the values shown.
Check every equation:
$2 + 2 = 4$ [both sides = 4]
$2 = 1 + 1$ [both sides = 2]
$1 = 2 \div 2$ [both sides = 1]

Solution - Exercise 54

(1) $A = C - B$ **A = 4**
(2) $2 \times C + B = 23$ **B = 5**
(3) $A = 4$ **C = 9**

Use equations 1, 3: $A = 4$; $C = B + 4$. Substitute into equation 2: $3 \times B + 8 = 23$. This gives B = 5; substitute back to obtain all the values shown.
Check every equation:
$4 = 9 - 5$ [both sides = 4]
$2 \times 9 + 5 = 23$ [both sides = 23]
$4 = 4$ [both sides = 4]

Solution - Exercise 55

(1) $B = 3 \times C$ **A = 8**
(2) $B + A = 11$ **B = 3**
(3) $C = A - 7$ **C = 1**

Use equations 1, 3: A = C + 7; B = 3 × C. Substitute into equation 2: 4 × C + 7 = 11. This gives C = 1; substitute back to obtain all the values shown.
Check every equation:
3 = 3 × 1 [both sides = 3]
3 + 8 = 11 [both sides = 11]
1 = 8 - 7 [both sides = 1]

Solution - Exercise 56

(1) $2 \times B + 2 \times C = A$ **A = 20**
(2) $4 \times C = B$ **B = 8**
(3) $B + C = 10$ **C = 2**

Use equations 2, 3: B = 8; C = 2. Substitute into equation 1: 20 = A. This gives A = 20; substitute back to obtain all the values shown.
Check every equation:
2 × 8 + 2 × 2 = 20 [both sides = 20]
4 × 2 = 8 [both sides = 8]
8 + 2 = 10 [both sides = 10]

Solution - Exercise 57

(1) $2 \times B = C$ **A = 12**
(2) $A \div 4 = B$ **B = 3**
(3) $15 - A = B$ **C = 6**

Use equations 2, 3: A = 12; B = 3. Substitute into equation 1: 6 = C. This gives C = 6; substitute back to obtain all the values shown.
Check every equation:
2 × 3 = 6 [both sides = 6]
12 ÷ 4 = 3 [both sides = 3]
15 - 12 = 3 [both sides = 3]

Solution - Exercise 58

(1) B = A + 6 **A = 13**
(2) B + A + C = 41 **B = 19**
(3) A = C + 4 **C = 9**

Use equations 1, 3: B = A + 6; C = A - 4. Substitute into equation 2: 3 × A + 2 = 41. This gives A = 13; substitute back to obtain all the values shown.
Check every equation:
19 = 13 + 6 [both sides = 19]
19 + 13 + 9 = 41 [both sides = 41]
13 = 9 + 4 [both sides = 13]

Solution - Exercise 59

(1) A + B = 24 **A = 6**
(2) A + C = 11 **B = 18**
(3) B + C = 23 **C = 5**

Use equations 1, 3: B = 24 - A; C = A - 1. Substitute into equation 2: 2 × A - 1 = 11. This gives A = 6; substitute back to obtain all the values shown.
Check every equation:
6 + 18 = 24 [both sides = 24]
6 + 5 = 11 [both sides = 11]
18 + 5 = 23 [both sides = 23]

Solution - Exercise 60

(1) A = C - 2 **A = 2**
(2) C = 2 × B **B = 2**
(3) B + A = 4 **C = 4**

Use equations 1, 3: A = 4 - B; C = 6 - B. Substitute into equation 2: 6 - B = 2 × B. This gives B = 2; substitute back to obtain all the values shown.
Check every equation:
2 = 4 - 2 [both sides = 2]
4 = 2 × 2 [both sides = 4]
2 + 2 = 4 [both sides = 4]

Solution - Exercise 61

(1)	C = A + 3	**A = 6**
(2)	A = B ÷ 2	**B = 12**
(3)	B + C = 21	**C = 9**

Use equations 1, 2: B = 2 × A; C = A + 3. Substitute into equation 3: 3 × A + 3 = 21. This gives A = 6; substitute back to obtain all the values shown.

Check every equation:

9 = 6 + 3 [both sides = 9]
6 = 12 ÷ 2 [both sides = 6]
12 + 9 = 21 [both sides = 21]

Solution - Exercise 62

(1)	4 × A + B = 52	**A = 12**
(2)	C = A - B	**B = 4**
(3)	C = 8	**C = 8**

Use equations 2, 3: B = A - 8; C = 8. Substitute into equation 1: 5 × A - 8 = 52. This gives A = 12; substitute back to obtain all the values shown.

Check every equation:

4 × 12 + 4 = 52 [both sides = 52]
8 = 12 - 4 [both sides = 8]
8 = 8 [both sides = 8]

Solution - Exercise 63

(1)	A = 2 × C	**A = 8**
(2)	A + B = 19	**B = 11**
(3)	C = B - 7	**C = 4**

Use equations 1, 3: A = 2 × C; B = C + 7. Substitute into equation 2: 3 × C + 7 = 19. This gives C = 4; substitute back to obtain all the values shown.

Check every equation:

8 = 2 × 4 [both sides = 8]
8 + 11 = 19 [both sides = 19]
4 = 11 - 7 [both sides = 4]

Solution - Exercise 64

(1) $C + A = 10$ **A = 2**
(2) $4 \times A = C$ **B = 18**
(3) $2 \times C + 1 \times A = B$ **C = 8**

Use equations 1, 2: A = 2; C = 8. Substitute into equation 3: 18 = B. This gives B = 18; substitute back to obtain all the values shown.
Check every equation:
8 + 2 = 10 [both sides = 10]
4 × 2 = 8 [both sides = 8]
2 × 8 + 1 × 2 = 18 [both sides = 18]

Solution - Exercise 65

(1) $12 - C = B$ **A = 12**
(2) $C \div 2 = B$ **B = 4**
(3) $3 \times B = A$ **C = 8**

Use equations 1, 2: B = 4; C = 8. Substitute into equation 3: 12 = A. This gives A = 12; substitute back to obtain all the values shown.
Check every equation:
12 - 8 = 4 [both sides = 4]
8 ÷ 2 = 4 [both sides = 4]
3 × 4 = 12 [both sides = 12]

Solution - Exercise 66

(1) $C + A + B = 42$ **A = 14**
(2) $A = B + 4$ **B = 10**
(3) $C = A + 4$ **C = 18**

Use equations 2, 3: B = A - 4; C = A + 4. Substitute into equation 1: 3 × A = 42. This gives A = 14; substitute back to obtain all the values shown.
Check every equation:
18 + 14 + 10 = 42 [both sides = 42]
14 = 10 + 4 [both sides = 14]
18 = 14 + 4 [both sides = 18]

Solution - Exercise 67

(1)	A + B = 19	**A = 14**
(2)	C + A = 26	**B = 5**
(3)	C + B = 17	**C = 12**

Use equations 2, 3: B = A - 9; C = 26 - A. Substitute into equation 1: 2 × A - 9 = 19. This gives A = 14; substitute back to obtain all the values shown.

Check every equation:

14 + 5 = 19 [both sides = 19]
12 + 14 = 26 [both sides = 26]
12 + 5 = 17 [both sides = 17]

Solution - Exercise 68

(1)	A = 2 × C	**A = 10**
(2)	C + B = 13	**B = 8**
(3)	B = A - 2	**C = 5**

Use equations 1, 3: A = 2 × C; B = 2 × C - 2. Substitute into equation 2: 3 × C - 2 = 13. This gives C = 5; substitute back to obtain all the values shown.

Check every equation:

10 = 2 × 5 [both sides = 10]
5 + 8 = 13 [both sides = 13]
8 = 10 - 2 [both sides = 8]

Solution - Exercise 69

(1)	B + C = 17	**A = 2**
(2)	C = A + 7	**B = 8**
(3)	A = B ÷ 4	**C = 9**

Use equations 2, 3: B = 4 × A; C = A + 7. Substitute into equation 1: 5 × A + 7 = 17. This gives A = 2; substitute back to obtain all the values shown.

Check every equation:

8 + 9 = 17 [both sides = 17]
9 = 2 + 7 [both sides = 9]
2 = 8 ÷ 4 [both sides = 2]

Solution - Exercise 70

(1) $4 \times C + A = 79$ **A = 15**
(2) $B = 1$ **B = 1**
(3) $B = C - A$ **C = 16**

Use equations 2, 3: B = 1; C = A + 1. Substitute into equation 1: 5 × A + 4 = 79. This gives A = 15; substitute back to obtain all the values shown.
Check every equation:
4 × 16 + 15 = 79 [both sides = 79]
1 = 1 [both sides = 1]
1 = 16 - 15 [both sides = 1]

Solution - Exercise 71

(1) $B + C = 6$ **A = 1**
(2) $B = 2 \times A$ **B = 2**
(3) $A = C - 3$ **C = 4**

Use equations 2, 3: B = 2 × A; C = A + 3. Substitute into equation 1: 3 × A + 3 = 6. This gives A = 1; substitute back to obtain all the values shown.
Check every equation:
2 + 4 = 6 [both sides = 6]
2 = 2 × 1 [both sides = 2]
1 = 4 - 3 [both sides = 1]

Solution - Exercise 72

(1) $C + B = 5$ **A = 5**
(2) $1 \times C + 1 \times B = A$ **B = 1**
(3) $4 \times B = C$ **C = 4**

Use equations 1, 3: B = 1; C = 4. Substitute into equation 2: 5 = A. This gives A = 5; substitute back to obtain all the values shown.
Check every equation:
4 + 1 = 5 [both sides = 5]
1 × 4 + 1 × 1 = 5 [both sides = 5]
4 × 1 = 4 [both sides = 4]

Solution - Exercise 73

(1)	A ÷ 3 = C	**A = 9**
(2)	2 × C = B	**B = 6**
(3)	12 - A = C	**C = 3**

Use equations 1, 3: A = 9; C = 3. Substitute into equation 2: 6 = B. This gives B = 6; substitute back to obtain all the values shown.

Check every equation:

9 ÷ 3 = 3 [both sides = 3]
2 × 3 = 6 [both sides = 6]
12 - 9 = 3 [both sides = 3]

Solution - Exercise 74

(1)	B = A + 1	**A = 9**
(2)	B + A + C = 23	**B = 10**
(3)	A = C + 5	**C = 4**

Use equations 1, 3: B = A + 1; C = A - 5. Substitute into equation 2: 3 × A - 4 = 23. This gives A = 9; substitute back to obtain all the values shown.

Check every equation:

10 = 9 + 1 [both sides = 10]
10 + 9 + 4 = 23 [both sides = 23]
9 = 4 + 5 [both sides = 9]

Solution - Exercise 75

(1)	C + B = 36	**A = 11**
(2)	A + B = 28	**B = 17**
(3)	A + C = 30	**C = 19**

Use equations 1, 3: B = A + 6; C = 30 - A. Substitute into equation 2: 2 × A + 6 = 28. This gives A = 11; substitute back to obtain all the values shown.

Check every equation:

19 + 17 = 36 [both sides = 36]
11 + 17 = 28 [both sides = 28]
11 + 19 = 30 [both sides = 30]

Solution - Exercise 76

(1)	B + C = 7	**A = 8**
(2)	A = 4 × B	**B = 2**
(3)	C = A - 3	**C = 5**

Use equations 2, 3: A = 4 × B; C = 4 × B - 3. Substitute into equation 1: 5 × B - 3 = 7. This gives B = 2; substitute back to obtain all the values shown.

Check every equation:

2 + 5 = 7 [both sides = 7]
8 = 4 × 2 [both sides = 8]
5 = 8 - 3 [both sides = 5]

Solution - Exercise 77

(1)	A + C = 14	**A = 8**
(2)	B = A ÷ 2	**B = 4**
(3)	C = B + 2	**C = 6**

Use equations 2, 3: A = 2 × B; C = B + 2. Substitute into equation 1: 3 × B + 2 = 14. This gives B = 4; substitute back to obtain all the values shown.

Check every equation:

8 + 6 = 14 [both sides = 14]
4 = 8 ÷ 2 [both sides = 4]
6 = 4 + 2 [both sides = 6]

Solution - Exercise 78

(1)	C = 10	**A = 12**
(2)	C = A - B	**B = 2**
(3)	4 × A + B = 50	**C = 10**

Use equations 1, 2: B = A - 10; C = 10. Substitute into equation 3: 5 × A - 10 = 50. This gives A = 12; substitute back to obtain all the values shown.

Check every equation:

10 = 10 [both sides = 10]
10 = 12 - 2 [both sides = 10]
4 × 12 + 2 = 50 [both sides = 50]

Solution - Exercise 79

(1) $A = 4 \times C$ **A = 12**
(2) $A + B = 19$ **B = 7**
(3) $C = B - 4$ **C = 3**

Use equations 1, 3: A = 4 × C; B = C + 4. Substitute into equation 2: 5 × C + 4 = 19. This gives C = 3; substitute back to obtain all the values shown.
Check every equation:
12 = 4 × 3 [both sides = 12]
12 + 7 = 19 [both sides = 19]
3 = 7 - 4 [both sides = 3]

Solution - Exercise 80

(1) $1 \times B + 2 \times C = A$ **A = 4**
(2) $B + C = 3$ **B = 2**
(3) $2 \times C = B$ **C = 1**

Use equations 2, 3: B = 2; C = 1. Substitute into equation 1: 4 = A. This gives A = 4; substitute back to obtain all the values shown.
Check every equation:
1 × 2 + 2 × 1 = 4 [both sides = 4]
2 + 1 = 3 [both sides = 3]
2 × 1 = 2 [both sides = 2]

Solution - Exercise 81

(1) $16 - C = B$ **A = 8**
(2) $C \div 3 = B$ **B = 4**
(3) $2 \times B = A$ **C = 12**

Use equations 1, 2: B = 4; C = 12. Substitute into equation 3: 8 = A. This gives A = 8; substitute back to obtain all the values shown.
Check every equation:
16 - 12 = 4 [both sides = 4]
12 ÷ 3 = 4 [both sides = 4]
2 × 4 = 8 [both sides = 8]

Solution - Exercise 82

(1) B = A + 2 **A = 13**
(2) B + A + C = 37 **B = 15**
(3) A = C + 4 **C = 9**

Use equations 1, 3: B = A + 2; C = A - 4. Substitute into equation 2: 3 × A - 2 = 37. This gives A = 13; substitute back to obtain all the values shown.

Check every equation:

15 = 13 + 2 [both sides = 15]
15 + 13 + 9 = 37 [both sides = 37]
13 = 9 + 4 [both sides = 13]

Solution - Exercise 83

(1) B + C = 14 **A = 5**
(2) A + C = 15 **B = 4**
(3) B + A = 9 **C = 10**

Use equations 1, 3: B = 9 - A; C = A + 5. Substitute into equation 2: 2 × A + 5 = 15. This gives A = 5; substitute back to obtain all the values shown.

Check every equation:

4 + 10 = 14 [both sides = 14]
5 + 10 = 15 [both sides = 15]
4 + 5 = 9 [both sides = 9]

Solution - Exercise 84

(1) C = A - 2 **A = 10**
(2) A = 2 × B **B = 5**
(3) B + C = 13 **C = 8**

Use equations 1, 2: A = 2 × B; C = 2 × B - 2. Substitute into equation 3: 3 × B - 2 = 13. This gives B = 5; substitute back to obtain all the values shown.

Check every equation:

8 = 10 - 2 [both sides = 8]
10 = 2 × 5 [both sides = 10]
5 + 8 = 13 [both sides = 13]

Solution - Exercise 85

(1) B = C + 1 **A = 4**
(2) C = A ÷ 4 **B = 2**
(3) A + B = 6 **C = 1**

Use equations 1, 2: A = 4 × C; B = C + 1. Substitute into equation 3: 5 × C + 1 = 6. This gives C = 1; substitute back to obtain all the values shown.
Check every equation:
2 = 1 + 1 [both sides = 2]
1 = 4 ÷ 4 [both sides = 1]
4 + 2 = 6 [both sides = 6]

Solution - Exercise 86

(1) C = A - B **A = 14**
(2) C = 6 **B = 8**
(3) 2 × A + B = 36 **C = 6**

Use equations 1, 2: B = A - 6; C = 6. Substitute into equation 3: 3 × A - 6 = 36. This gives A = 14; substitute back to obtain all the values shown.
Check every equation:
6 = 14 - 8 [both sides = 6]
6 = 6 [both sides = 6]
2 × 14 + 8 = 36 [both sides = 36]

Solution - Exercise 87

(1) B + A = 31 **A = 11**
(2) C = A - 6 **B = 20**
(3) B = 4 × C **C = 5**

Use equations 2, 3: A = C + 6; B = 4 × C. Substitute into equation 1: 5 × C + 6 = 31. This gives C = 5; substitute back to obtain all the values shown.
Check every equation:
20 + 11 = 31 [both sides = 31]
5 = 11 - 6 [both sides = 5]
20 = 4 × 5 [both sides = 20]

Solution - Exercise 88

(1) $4 \times A = B$ **A = 3**
(2) $B + A = 15$ **B = 12**
(3) $1 \times B + 1 \times A = C$ **C = 15**

Use equations 1, 2: A = 3; B = 12. Substitute into equation 3: 15 = C. This gives C = 15; substitute back to obtain all the values shown.
Check every equation:
4 × 3 = 12 [both sides = 12]
12 + 3 = 15 [both sides = 15]
1 × 12 + 1 × 3 = 15 [both sides = 15]

Solution - Exercise 89

(1) $6 - A = B$ **A = 4**
(2) $2 \times B = C$ **B = 2**
(3) $A \div 2 = B$ **C = 4**

Use equations 1, 3: A = 4; B = 2. Substitute into equation 2: 4 = C. This gives C = 4; substitute back to obtain all the values shown.
Check every equation:
6 - 4 = 2 [both sides = 2]
2 × 2 = 4 [both sides = 4]
4 ÷ 2 = 2 [both sides = 2]

Solution - Exercise 90

(1) $A = B + 6$ **A = 19**
(2) $A + B + C = 40$ **B = 13**
(3) $B = C + 5$ **C = 8**

Use equations 1, 3: A = B + 6; C = B - 5. Substitute into equation 2: 3 × B + 1 = 40. This gives B = 13; substitute back to obtain all the values shown.
Check every equation:
19 = 13 + 6 [both sides = 19]
19 + 13 + 8 = 40 [both sides = 40]
13 = 8 + 5 [both sides = 13]

Solution - Exercise 91

(1) C + B = 30 **A = 10**
(2) A + C = 22 **B = 18**
(3) A + B = 28 **C = 12**

Use equations 1, 3: B = 28 - A; C = A + 2. Substitute into equation 2: 2 × A + 2 = 22. This gives A = 10; substitute back to obtain all the values shown.
Check every equation:
12 + 18 = 30 [both sides = 30]
10 + 12 = 22 [both sides = 22]
10 + 18 = 28 [both sides = 28]

Solution - Exercise 92

(1) C + A = 14 **A = 10**
(2) B = 4 × C **B = 16**
(3) A = B - 6 **C = 4**

Use equations 2, 3: A = 4 × C - 6; B = 4 × C. Substitute into equation 1: 5 × C - 6 = 14. This gives C = 4; substitute back to obtain all the values shown.
Check every equation:
4 + 10 = 14 [both sides = 14]
16 = 4 × 4 [both sides = 16]
10 = 16 - 6 [both sides = 10]

Solution - Exercise 93

(1) B = C ÷ 4 **A = 8**
(2) A = B + 6 **B = 2**
(3) C + A = 16 **C = 8**

Use equations 1, 2: A = B + 6; C = 4 × B. Substitute into equation 3: 5 × B + 6 = 16. This gives B = 2; substitute back to obtain all the values shown.
Check every equation:
2 = 8 ÷ 4 [both sides = 2]
8 = 2 + 6 [both sides = 8]
8 + 8 = 16 [both sides = 16]

Solution - Exercise 94

(1)	4 × C = A	**A = 12**
(2)	3 × C = D	**B = 9**
(3)	D - C + A - B = 9	**C = 3**
(4)	6 + C = B	**D = 9**

Use equations 1, 2, 4: A = 4 × C; B = C + 6; D = 3 × C. Substitute into equation 3: 5 × C - 6 = 9. This gives C = 3; substitute back to obtain all the values shown.

Check every equation:

4 × 3 = 12 [both sides = 12]
3 × 3 = 9 [both sides = 9]
9 - 3 + 12 - 9 = 9 [both sides = 9]
6 + 3 = 9 [both sides = 9]

Solution - Exercise 95

(1)	C + D - B = 19	**A = 14**
(2)	3 × C - 1 = A	**B = 6**
(3)	11 - C = B	**C = 5**
(4)	4 × C = D	**D = 20**

Use equations 1, 3, 4: B = 6; C = 5; D = 20. Substitute into equation 2: 14 = A. This gives A = 14; substitute back to obtain all the values shown.

Check every equation:

5 + 20 - 6 = 19 [both sides = 19]
3 × 5 - 1 = 14 [both sides = 14]
11 - 5 = 6 [both sides = 6]
4 × 5 = 20 [both sides = 20]

Solution - Exercise 96

(1)	D = 2 × C - 1	**A = 1**
(2)	A = 2 × B - 3	**B = 2**
(3)	C = A + 1	**C = 2**
(4)	D - C + A + B = 4	**D = 3**

Use equations 1, 3, 4: B = 4 - 2 × A; C = A + 1; D = 2 × A + 1. Substitute into equation 2: A = 5 - 4 × A. This gives A = 1; substitute back to obtain all the values shown.

Check every equation:

3 = 2 × 2 - 1 [both sides = 3]
1 = 2 × 2 - 3 [both sides = 1]
2 = 1 + 1 [both sides = 2]
3 - 2 + 1 + 2 = 4 [both sides = 4]

Solution - Exercise 97

(1)	B + A = 6	**A = 2**
(2)	C + B = 16	**B = 4**
(3)	D = 3 × A	**C = 12**
(4)	D + C = 18	**D = 6**

Use equations 1, 3, 4: A = 6 - B; C = 3 × B; D = 18 - 3 × B. Substitute into equation 2: 4 × B = 16. This gives B = 4; substitute back to obtain all the values shown.

Check every equation:

4 + 2 = 6 [both sides = 6]
12 + 4 = 16 [both sides = 16]
6 = 3 × 2 [both sides = 6]
6 + 12 = 18 [both sides = 18]

Solution - Exercise 98

(1)	A ÷ 3 + D = B	**A = 12**
(2)	D = C + 2	**B = 10**
(3)	C = B - D	**C = 4**
(4)	A + D + B + C = 32	**D = 6**

Use equations 1, 2, 3: A = 3 × C; B = 2 × C + 2; D = C + 2. Substitute into equation 4: 7 × C + 4 = 32. This gives C = 4; substitute back to obtain all the values shown.

Check every equation:

12 ÷ 3 + 6 = 10 [both sides = 10]
6 = 4 + 2 [both sides = 6]
4 = 10 - 6 [both sides = 4]
12 + 6 + 10 + 4 = 32 [both sides = 32]

Solution - Exercise 99

(1)	5 × A - D = B	**A = 3**
(2)	A + C = 13	**B = 13**
(3)	D + B = 15	**C = 10**
(4)	C = B - A	**D = 2**

Use equations 1, 3, 4: A = 3; C = B - 3; D = 15 - B. Substitute into equation 2: B = 13. This gives B = 13; substitute back to obtain all the values shown.

Check every equation:

5 × 3 - 2 = 13 [both sides = 13]
3 + 10 = 13 [both sides = 13]
2 + 13 = 15 [both sides = 15]
10 = 13 - 3 [both sides = 10]

Solution - Exercise 100

(1) C + B + D = 24 **A = 3**
(2) D = 2 × A **B = 6**
(3) C - B = 6 **C = 12**
(4) B = A + 3 **D = 6**

Use equations 2, 3, 4: A = B - 3; C = B + 6; D = 2 × B - 6. Substitute into equation 1: 4 × B = 24. This gives B = 6; substitute back to obtain all the values shown.
Check every equation:
12 + 6 + 6 = 24 [both sides = 24]
6 = 2 × 3 [both sides = 6]
12 - 6 = 6 [both sides = 6]
6 = 3 + 3 [both sides = 6]

Solution - Exercise 101

(1) B = 5 × C - 5 **A = 7**
(2) D = 4 × C - 4 **B = 5**
(3) A = C + 5 **C = 2**
(4) B + C = A + D - 4 **D = 4**

Use equations 2, 3, 4: A = C + 5; B = 4 × C - 3; D = 4 × C - 4. Substitute into equation 1: 4 × C - 3 = 5 × C - 5. This gives C = 2; substitute back to obtain all the values shown.
Check every equation:
5 = 5 × 2 - 5 [both sides = 5]
4 = 4 × 2 - 4 [both sides = 4]
7 = 2 + 5 [both sides = 7]
5 + 2 = 7 + 4 - 4 [both sides = 7]

Solution - Exercise 102

(1) C - A + D - B = 21 **A = 5**
(2) 4 × A = D **B = 14**
(3) 9 + A = B **C = 20**
(4) 4 × A = C **D = 20**

Use equations 2, 3, 4: B = A + 9; C = 4 × A; D = 4 × A. Substitute into equation 1: 6 × A - 9 = 21. This gives A = 5; substitute back to obtain all the values shown.
Check every equation:
20 - 5 + 20 - 14 = 21 [both sides = 21]
4 × 5 = 20 [both sides = 20]
9 + 5 = 14 [both sides = 14]
4 × 5 = 20 [both sides = 20]

Solution - Exercise 103

(1)	5 × C - 1 = B	**A = 8**
(2)	10 - C = A	**B = 9**
(3)	5 × C = D	**C = 2**
(4)	C + D - A = 4	**D = 10**

Use equations 2, 3, 4: A = 8; C = 2; D = 10. Substitute into equation 1: 9 = B. This gives B = 9; substitute back to obtain all the values shown.

Check every equation:

5 × 2 - 1 = 9 [both sides = 9]
10 - 2 = 8 [both sides = 8]
5 × 2 = 10 [both sides = 10]
2 + 10 - 8 = 4 [both sides = 4]

Solution - Exercise 104

(1)	A - B + C + D = 19	**A = 17**
(2)	A = 3 × B - 4	**B = 7**
(3)	C = 3 × D - 7	**C = 5**
(4)	B = C + 2	**D = 4**

Use equations 1, 2, 4: A = 3 × C + 2; B = C + 2; D = 19 - 3 × C. Substitute into equation 3: C = 50 - 9 × C. This gives C = 5; substitute back to obtain all the values shown.

Check every equation:

17 - 7 + 5 + 4 = 19 [both sides = 19]
17 = 3 × 7 - 4 [both sides = 17]
5 = 3 × 4 - 7 [both sides = 5]
7 = 5 + 2 [both sides = 7]

Solution - Exercise 105

(1)	D + C = 12	**A = 6**
(2)	B + D = 11	**B = 2**
(3)	A = 2 × C	**C = 3**
(4)	A + B = 8	**D = 9**

Use equations 1, 2, 3: A = 2 × C; B = C - 1; D = 12 - C. Substitute into equation 4: 3 × C - 1 = 8. This gives C = 3; substitute back to obtain all the values shown.

Check every equation:

9 + 3 = 12 [both sides = 12]
2 + 9 = 11 [both sides = 11]
6 = 2 × 3 [both sides = 6]
6 + 2 = 8 [both sides = 8]

Solution - Exercise 106

(1)	A + D + C + B = 29	**A = 15**
(2)	A ÷ 5 + D = C	**B = 3**
(3)	D = B + 1	**C = 7**
(4)	B = C - D	**D = 4**

Use equations 2, 3, 4: A = 5 × B; C = 2 × B + 1; D = B + 1. Substitute into equation 1: 9 × B + 2 = 29. This gives B = 3; substitute back to obtain all the values shown.

Check every equation:

15 + 4 + 7 + 3 = 29 [both sides = 29]
15 ÷ 5 + 4 = 7 [both sides = 7]
4 = 3 + 1 [both sides = 4]
3 = 7 - 4 [both sides = 3]

Solution - Exercise 107

(1)	D + B = 16	**A = 8**
(2)	4 × D - A = C	**B = 10**
(3)	B = C - D	**C = 16**
(4)	A + C = 24	**D = 6**

Use equations 2, 3, 4: A = 24 - C; B = C - 6; D = 6. Substitute into equation 1: C = 16. This gives C = 16; substitute back to obtain all the values shown.

Check every equation:

6 + 10 = 16 [both sides = 16]
4 × 6 - 8 = 16 [both sides = 16]
10 = 16 - 6 [both sides = 10]
8 + 16 = 24 [both sides = 24]

Solution - Exercise 108

(1)	A = 3 × D	**A = 15**
(2)	B = D + 3	**B = 8**
(3)	C + B + A = 33	**C = 10**
(4)	C - B = 2	**D = 5**

Use equations 1, 2, 4: A = 3 × D; B = D + 3; C = D + 5. Substitute into equation 3: 5 × D + 8 = 33. This gives D = 5; substitute back to obtain all the values shown.

Check every equation:

15 = 3 × 5 [both sides = 15]
8 = 5 + 3 [both sides = 8]
10 + 8 + 15 = 33 [both sides = 33]
10 - 8 = 2 [both sides = 2]

Solution - Exercise 109

(1)	A = 4 × C - 3	**A = 17**
(2)	D = C + 2	**B = 15**
(3)	B = 5 × C - 10	**C = 5**
(4)	A + C = D + B	**D = 7**

Use equations 1, 3, 4: A = 4 × C - 3; B = 5 × C - 10; D = 7. Substitute into equation 2: 7 = C + 2. This gives C = 5; substitute back to obtain all the values shown.

Check every equation:

17 = 4 × 5 - 3 [both sides = 17]
7 = 5 + 2 [both sides = 7]
15 = 5 × 5 - 10 [both sides = 15]
17 + 5 = 7 + 15 [both sides = 22]

Solution - Exercise 110

(1)	2 × A = C	**A = 4**
(2)	3 + A = D	**B = 16**
(3)	C - A + B - D = 13	**C = 8**
(4)	4 × A = B	**D = 7**

Use equations 1, 2, 3: B = 16; C = 2 × A; D = A + 3. Substitute into equation 4: 4 × A = 16. This gives A = 4; substitute back to obtain all the values shown.

Check every equation:

2 × 4 = 8 [both sides = 8]
3 + 4 = 7 [both sides = 7]
8 - 4 + 16 - 7 = 13 [both sides = 13]
4 × 4 = 16 [both sides = 16]

Solution - Exercise 111

(1)	11 - B = C	**A = 9**
(2)	5 × B = D	**B = 2**
(3)	5 × B - 1 = A	**C = 9**
(4)	B + D - C = 3	**D = 10**

Use equations 1, 2, 4: B = 2; C = 9; D = 10. Substitute into equation 3: 9 = A. This gives A = 9; substitute back to obtain all the values shown.

Check every equation:

11 - 2 = 9 [both sides = 9]
5 × 2 = 10 [both sides = 10]
5 × 2 - 1 = 9 [both sides = 9]
2 + 10 - 9 = 3 [both sides = 3]

Solution - Exercise 112

(1)	C = 5 × B - 6	**A = 17**
(2)	A - D + C + B = 10	**B = 2**
(3)	D = C + 9	**C = 4**
(4)	A = 2 × D - 9	**D = 13**

Use equations 2, 3, 4: A = 2 × C + 9; B = 10 - 2 × C; D = C + 9. Substitute into equation 1: C = 44 - 10 × C. This gives C = 4; substitute back to obtain all the values shown.

Check every equation:

4 = 5 × 2 - 6 [both sides = 4]
17 - 13 + 4 + 2 = 10 [both sides = 10]
13 = 4 + 9 [both sides = 13]
17 = 2 × 13 - 9 [both sides = 17]

Solution - Exercise 113

(1)	C + A = 24	**A = 13**
(2)	B = 4 × D	**B = 4**
(3)	B + C = 15	**C = 11**
(4)	A + D = 14	**D = 1**

Use equations 1, 3, 4: A = B + 9; C = 15 - B; D = 5 - B. Substitute into equation 2: B = 20 - 4 × B. This gives B = 4; substitute back to obtain all the values shown.

Check every equation:

11 + 13 = 24 [both sides = 24]
4 = 4 × 1 [both sides = 4]
4 + 11 = 15 [both sides = 15]
13 + 1 = 14 [both sides = 14]

Solution - Exercise 114

(1)	B ÷ 2 + C = A	**A = 13**
(2)	B + C + A + D = 34	**B = 8**
(3)	C = D + 5	**C = 9**
(4)	D = A - C	**D = 4**

Use equations 1, 2, 3: A = 13; B = 16 - 2 × D; C = D + 5. Substitute into equation 4: D = 8 - D. This gives D = 4; substitute back to obtain all the values shown.

Check every equation:

8 ÷ 2 + 9 = 13 [both sides = 13]
8 + 9 + 13 + 4 = 34 [both sides = 34]
9 = 4 + 5 [both sides = 9]
4 = 13 - 9 [both sides = 4]

Solution - Exercise 115

(1)	D + B = 20	**A = 4**
(2)	B = C - D	**B = 14**
(3)	A + C = 24	**C = 20**
(4)	4 × D - A = C	**D = 6**

Use equations 2, 3, 4: A = 24 - C; B = C - 6; D = 6. Substitute into equation 1: C = 20. This gives C = 20; substitute back to obtain all the values shown.

Check every equation:

6 + 14 = 20 [both sides = 20]
14 = 20 - 6 [both sides = 14]
4 + 20 = 24 [both sides = 24]
4 × 6 - 4 = 20 [both sides = 20]

Solution - Exercise 116

(1)	C = 4 × A	**A = 2**
(2)	B = A + 5	**B = 7**
(3)	D - B = 1	**C = 8**
(4)	D + B + C = 23	**D = 8**

Use equations 1, 2, 3: B = A + 5; C = 4 × A; D = A + 6. Substitute into equation 4: 6 × A + 11 = 23. This gives A = 2; substitute back to obtain all the values shown.

Check every equation:

8 = 4 × 2 [both sides = 8]
7 = 2 + 5 [both sides = 7]
8 - 7 = 1 [both sides = 1]
8 + 7 + 8 = 23 [both sides = 23]

Solution - Exercise 117

(1)	D = 2 × B - 7	**A = 11**
(2)	A = B + 5	**B = 6**
(3)	C = 4 × B - 5	**C = 19**
(4)	D + B = A + C - 19	**D = 5**

Use equations 1, 3, 4: B = 17 - A; C = 63 - 4 × A; D = 27 - 2 × A. Substitute into equation 2: A = 22 - A. This gives A = 11; substitute back to obtain all the values shown.

Check every equation:

5 = 2 × 6 - 7 [both sides = 5]
11 = 6 + 5 [both sides = 11]
19 = 4 × 6 - 5 [both sides = 19]
5 + 6 = 11 + 19 - 19 [both sides = 11]

Solution - Exercise 118

(1)	D - A + B - C = 14	**A = 3**
(2)	4 × A = D	**B = 9**
(3)	3 × A = B	**C = 4**
(4)	1 + A = C	**D = 12**

Use equations 2, 3, 4: B = 3 × A; C = A + 1; D = 4 × A. Substitute into equation 1: 5 × A - 1 = 14. This gives A = 3; substitute back to obtain all the values shown.

Check every equation:

12 - 3 + 9 - 4 = 14 [both sides = 14]
4 × 3 = 12 [both sides = 12]
3 × 3 = 9 [both sides = 9]
1 + 3 = 4 [both sides = 4]

Solution - Exercise 119

(1)	C + A - B = 6	**A = 9**
(2)	2 × C - 1 = D	**B = 6**
(3)	9 - C = B	**C = 3**
(4)	3 × C = A	**D = 5**

Use equations 1, 3, 4: A = 9; B = 6; C = 3. Substitute into equation 2: 5 = D. This gives D = 5; substitute back to obtain all the values shown.

Check every equation:

3 + 9 - 6 = 6 [both sides = 6]
2 × 3 - 1 = 5 [both sides = 5]
9 - 3 = 6 [both sides = 6]
3 × 3 = 9 [both sides = 9]

Solution - Exercise 120

(1)	A = 3 × D - 3	**A = 6**
(2)	D = C + 1	**B = 2**
(3)	C = 3 × B - 4	**C = 2**
(4)	A - D + C + B = 7	**D = 3**

Use equations 1, 2, 4: A = 3 × C; B = 8 - 3 × C; D = C + 1. Substitute into equation 3: C = 20 - 9 × C. This gives C = 2; substitute back to obtain all the values shown.

Check every equation:

6 = 3 × 3 - 3 [both sides = 6]
3 = 2 + 1 [both sides = 3]
2 = 3 × 2 - 4 [both sides = 2]
6 - 3 + 2 + 2 = 7 [both sides = 7]

Solution - Exercise 121

(1)	D + A = 10	**A = 3**
(2)	C + D = 11	**B = 12**
(3)	B + C = 16	**C = 4**
(4)	B = 4 × A	**D = 7**

Use equations 1, 2, 4: B = 4 × A; C = A + 1; D = 10 - A. Substitute into equation 3: 5 × A + 1 = 16. This gives A = 3; substitute back to obtain all the values shown.

Check every equation:

7 + 3 = 10 [both sides = 10]
4 + 7 = 11 [both sides = 11]
12 + 4 = 16 [both sides = 16]
12 = 4 × 3 [both sides = 12]

Solution - Exercise 122

(1)	A = C - B	**A = 2**
(2)	D ÷ 3 + B = C	**B = 9**
(3)	D + B + C + A = 28	**C = 11**
(4)	B = A + 7	**D = 6**

Use equations 1, 2, 4: B = A + 7; C = 2 × A + 7; D = 3 × A. Substitute into equation 3: 7 × A + 14 = 28. This gives A = 2; substitute back to obtain all the values shown.

Check every equation:

2 = 11 - 9 [both sides = 2]
6 ÷ 3 + 9 = 11 [both sides = 11]
6 + 9 + 11 + 2 = 28 [both sides = 28]
9 = 2 + 7 [both sides = 9]

Solution - Exercise 123

(1)	D = B - C	**A = 7**
(2)	C + D = 13	**B = 13**
(3)	2 × C - A = B	**C = 10**
(4)	A + B = 20	**D = 3**

Use equations 2, 3, 4: B = 20 - A; C = 10; D = 3. Substitute into equation 1: 3 = 10 - A. This gives A = 7; substitute back to obtain all the values shown.

Check every equation:

3 = 13 - 10 [both sides = 3]
10 + 3 = 13 [both sides = 13]
2 × 10 - 7 = 13 [both sides = 13]
7 + 13 = 20 [both sides = 20]

Solution - Exercise 124

(1)	A = D + 1	**A = 4**
(2)	B = 3 × D	**B = 9**
(3)	C + A + B = 24	**C = 11**
(4)	C - A = 7	**D = 3**

Use equations 1, 2, 4: A = D + 1; B = 3 × D; C = D + 8. Substitute into equation 3: 5 × D + 9 = 24. This gives D = 3; substitute back to obtain all the values shown.

Check every equation:

4 = 3 + 1 [both sides = 4]
9 = 3 × 3 [both sides = 9]
11 + 4 + 9 = 24 [both sides = 24]
11 - 4 = 7 [both sides = 7]

Solution - Exercise 125

(1)	B = 3 × A - 4	**A = 5**
(2)	D = A + 2	**B = 11**
(3)	C = 4 × A - 5	**C = 15**
(4)	C + A = D + B + 2	**D = 7**

Use equations 1, 2, 4: B = 3 × A - 4; C = 3 × A; D = A + 2. Substitute into equation 3: 3 × A = 4 × A - 5. This gives A = 5; substitute back to obtain all the values shown.

Check every equation:

11 = 3 × 5 - 4 [both sides = 11]
7 = 5 + 2 [both sides = 7]
15 = 4 × 5 - 5 [both sides = 15]
15 + 5 = 7 + 11 + 2 [both sides = 20]

Solution - Exercise 126

(1)	3 × D = A	**A = 9**
(2)	3 × D = C	**B = 6**
(3)	C - D + A - B = 9	**C = 9**
(4)	3 + D = B	**D = 3**

Use equations 1, 2, 4: A = 3 × D; B = D + 3; C = 3 × D. Substitute into equation 3: 4 × D - 3 = 9. This gives D = 3; substitute back to obtain all the values shown.

Check every equation:

3 × 3 = 9 [both sides = 9]
3 × 3 = 9 [both sides = 9]
9 - 3 + 9 - 6 = 9 [both sides = 9]
3 + 3 = 6 [both sides = 6]

Solution - Exercise 127

(1)	4 × C = B	**A = 9**
(2)	3 × C - 1 = D	**B = 12**
(3)	12 - C = A	**C = 3**
(4)	C + B - A = 6	**D = 8**

Use equations 1, 3, 4: A = 9; B = 12; C = 3. Substitute into equation 2: 8 = D. This gives D = 8; substitute back to obtain all the values shown.

Check every equation:

4 × 3 = 12 [both sides = 12]
3 × 3 - 1 = 8 [both sides = 8]
12 - 3 = 9 [both sides = 9]
3 + 12 - 9 = 6 [both sides = 6]

Solution - Exercise 128

(1)	D = 4 × B - 2	**A = 7**
(2)	A = 3 × C - 5	**B = 1**
(3)	A - C + D + B = 6	**C = 4**
(4)	C = D + 2	**D = 2**

Use equations 1, 2, 4: A = 12 × B - 5; C = 4 × B; D = 4 × B - 2. Substitute into equation 3: 13 × B - 7 = 6. This gives B = 1; substitute back to obtain all the values shown.

Check every equation:

2 = 4 × 1 - 2 [both sides = 2]
7 = 3 × 4 - 5 [both sides = 7]
7 - 4 + 2 + 1 = 6 [both sides = 6]
4 = 2 + 2 [both sides = 4]

Solution - Exercise 129

(1)	D = 4 × B	**A = 11**
(2)	C + B = 15	**B = 5**
(3)	A + C = 21	**C = 10**
(4)	D + A = 31	**D = 20**

Use equations 1, 2, 3: A = B + 6; C = 15 - B; D = 4 × B. Substitute into equation 4: 5 × B + 6 = 31. This gives B = 5; substitute back to obtain all the values shown.

Check every equation:

20 = 4 × 5 [both sides = 20]
10 + 5 = 15 [both sides = 15]
11 + 10 = 21 [both sides = 21]
20 + 11 = 31 [both sides = 31]

Solution - Exercise 130

(1)	B + C + A + D = 31	**A = 11**
(2)	D = A - C	**B = 9**
(3)	C = D + 5	**C = 8**
(4)	B ÷ 3 + C = A	**D = 3**

Use equations 2, 3, 4: A = 2 × D + 5; B = 3 × D; C = D + 5. Substitute into equation 1: 7 × D + 10 = 31. This gives D = 3; substitute back to obtain all the values shown.

Check every equation:

9 + 8 + 11 + 3 = 31 [both sides = 31]
3 = 11 - 8 [both sides = 3]
8 = 3 + 5 [both sides = 8]
9 ÷ 3 + 8 = 11 [both sides = 11]

Solution - Exercise 131

(1)	B = D - C	**A = 8**
(2)	A + D = 16	**B = 4**
(3)	C + B = 8	**C = 4**
(4)	4 × C - A = D	**D = 8**

Use equations 2, 3, 4: A = 16 - D; B = 4; C = 4. Substitute into equation 1: 4 = D - 4. This gives D = 8; substitute back to obtain all the values shown.

Check every equation:

4 = 8 - 4 [both sides = 4]
8 + 8 = 16 [both sides = 16]
4 + 4 = 8 [both sides = 8]
4 × 4 - 8 = 8 [both sides = 8]

Solution - Exercise 132

(1)	C = A + 2	**A = 3**
(2)	D = 4 × A	**B = 13**
(3)	B - C = 8	**C = 5**
(4)	B + C + D = 30	**D = 12**

Use equations 1, 2, 3: A = C - 2; B = C + 8; D = 4 × C - 8. Substitute into equation 4: 6 × C = 30. This gives C = 5; substitute back to obtain all the values shown.

Check every equation:

5 = 3 + 2 [both sides = 5]
12 = 4 × 3 [both sides = 12]
13 - 5 = 8 [both sides = 8]
13 + 5 + 12 = 30 [both sides = 30]

Solution - Exercise 133

(1)	B = 4 × C - 3	**A = 9**
(2)	A = 5 × C - 6	**B = 9**
(3)	A + C = D + B - 3	**C = 3**
(4)	D = C + 3	**D = 6**

Use equations 1, 2, 3: A = 5 × C - 6; B = 4 × C - 3; D = 2 × C. Substitute into equation 4: 2 × C = C + 3. This gives C = 3; substitute back to obtain all the values shown.

Check every equation:

9 = 4 × 3 - 3 [both sides = 9]
9 = 5 × 3 - 6 [both sides = 9]
9 + 3 = 6 + 9 - 3 [both sides = 12]
6 = 3 + 3 [both sides = 6]

Solution - Exercise 134

(1)	6 + A = D	**A = 3**
(2)	B - A + C - D = 6	**B = 6**
(3)	4 × A = C	**C = 12**
(4)	2 × A = B	**D = 9**

Use equations 1, 2, 4: B = 2 × A; C = 12; D = A + 6. Substitute into equation 3: 4 × A = 12. This gives A = 3; substitute back to obtain all the values shown.

Check every equation:

6 + 3 = 9 [both sides = 9]
6 - 3 + 12 - 9 = 6 [both sides = 6]
4 × 3 = 12 [both sides = 12]
2 × 3 = 6 [both sides = 6]

Solution - Exercise 135

(1)	2 × A - 1 = D	**A = 4**
(2)	2 × A = C	**B = 9**
(3)	13 - A = B	**C = 8**
(4)	A + C - B = 3	**D = 7**

Use equations 2, 3, 4: A = 4; B = 9; C = 8. Substitute into equation 1: 7 = D. This gives D = 7; substitute back to obtain all the values shown.

Check every equation:

2 × 4 - 1 = 7 [both sides = 7]
2 × 4 = 8 [both sides = 8]
13 - 4 = 9 [both sides = 9]
4 + 8 - 9 = 3 [both sides = 3]

Solution - Exercise 136

(1)	$A = 2 \times B - 7$	**A = 7**
(2)	$A - B + D + C = 8$	**B = 7**
(3)	$B = D + 2$	**C = 3**
(4)	$D = 3 \times C - 4$	**D = 5**

Use equations 1, 2, 3: $A = 2 \times D - 3$; $B = D + 2$; $C = 13 - 2 \times D$. Substitute into equation 4: $D = 35 - 6 \times D$. This gives D = 5; substitute back to obtain all the values shown.

Check every equation:

$7 = 2 \times 7 - 7$ [both sides = 7]
$7 - 7 + 5 + 3 = 8$ [both sides = 8]
$7 = 5 + 2$ [both sides = 7]
$5 = 3 \times 3 - 4$ [both sides = 5]

Solution - Exercise 137

(1)	$B + C = 11$	**A = 1**
(2)	$C + A = 6$	**B = 6**
(3)	$D + B = 8$	**C = 5**
(4)	$D = 2 \times A$	**D = 2**

Use equations 1, 2, 4: $B = A + 5$; $C = 6 - A$; $D = 2 \times A$. Substitute into equation 3: $3 \times A + 5 = 8$. This gives A = 1; substitute back to obtain all the values shown.

Check every equation:

$6 + 5 = 11$ [both sides = 11]
$5 + 1 = 6$ [both sides = 6]
$2 + 6 = 8$ [both sides = 8]
$2 = 2 \times 1$ [both sides = 2]

Solution - Exercise 138

(1)	$D = A + 4$	**A = 3**
(2)	$C + D + B + A = 32$	**B = 10**
(3)	$C \div 4 + D = B$	**C = 12**
(4)	$A = B - D$	**D = 7**

Use equations 1, 3, 4: $B = 2 \times A + 4$; $C = 4 \times A$; $D = A + 4$. Substitute into equation 2: $8 \times A + 8 = 32$. This gives A = 3; substitute back to obtain all the values shown.

Check every equation:

$7 = 3 + 4$ [both sides = 7]
$12 + 7 + 10 + 3 = 32$ [both sides = 32]
$12 \div 4 + 7 = 10$ [both sides = 10]
$3 = 10 - 7$ [both sides = 3]

Solution - Exercise 139

(1)	5 × B - D = A	**A = 14**
(2)	D + A = 15	**B = 3**
(3)	C = A - B	**C = 11**
(4)	B + C = 14	**D = 1**

Use equations 1, 2, 3: B = 3; C = A - 3; D = 15 - A. Substitute into equation 4: A = 14. This gives A = 14; substitute back to obtain all the values shown.

Check every equation:

5 × 3 - 1 = 14 [both sides = 14]
1 + 14 = 15 [both sides = 15]
11 = 14 - 3 [both sides = 11]
3 + 11 = 14 [both sides = 14]

Solution - Exercise 140

(1)	C - D = 4	**A = 5**
(2)	D = A + 2	**B = 15**
(3)	B = 3 × A	**C = 11**
(4)	C + D + B = 33	**D = 7**

Use equations 1, 2, 3: B = 3 × A; C = A + 6; D = A + 2. Substitute into equation 4: 5 × A + 8 = 33. This gives A = 5; substitute back to obtain all the values shown.

Check every equation:

11 - 7 = 4 [both sides = 4]
7 = 5 + 2 [both sides = 7]
15 = 3 × 5 [both sides = 15]
11 + 7 + 15 = 33 [both sides = 33]